

This document is intended as a guide for the senior project team to assess its performance in a number of dimensions. Add additional items that you feel are appropriate.

This self-assessment will be one of multiple elements that your faculty coach uses to arrive at an assessment of the team's performance for this second term. The other elements that the faculty coach will use include: direct observation of the team, team peer evaluations, reviews by other faculty during the project presentation, sponsor evaluation, and project deliverables. These self-assessments will also be used as part of the SE program's accreditation and curriculum improvement efforts.

To complete this self-assessment the team should carefully consider each of the questions and provide an honest evaluation of the team's performance.

Product

1. Did the team prepare all the documentation artifacts requested by your faculty coach and sponsor? Were these documents carefully inspected prior to delivery? How would you assess the quality of the document artifacts?
 - a. The team prepared and provided all request documentation by both the faculty and sponsor. This includes all senior project deliverables and important documentation for handing off our project. As much as possible each member of the team reviewed these documents prior to sending them out to ensure the utmost quality. In general, the team agrees that deliverables have been satisfactory.
2. How well did the team elicit the requirements? What approaches were used to elicit the requirements? Were key requirements missed? What methodology was used to document and validate the project requirements?
 - a. Requirements were elicited from the sponsor over time as we became more familiar with the project domain during the first semester. Prior to the beginning of the second semester it was abundantly clear that the team needed to prioritize the completion of a polished MVP that can be beta tested by real users. As such, during one of our initial meetings this semester the team brainstormed to create a list of priorities that would drive the focus of our development. Afterwards, we communicated with the sponsor to come to an agreement on the goals for this semester and began development accordingly. While there was no clear use of methodology to elicit requirements this semester, the team stayed true to our agile scrum development process and used bi-weekly sprint demos/meetings to drive and validate decisions.

3. Did the team explore the entire design space before arriving at a final design? Have there been many errors found in the design? Was it necessary to make major changes to any part of the design? What were the reasons for the change?
 - a. A large portion of our initial design space was occupied by Shopify's proprietary development system. This includes data APIs to query, update, and delete Shopify entities such as selling plan groups as well as methods of listening to Shopify events like "on app delete". However, this semester as our AI and cloud features became more concrete their design needed to be more distinctly architected. For example, our team finalized the type of LLM model to be used with our churn score explanation feature and implemented hosting via AWS EC2 instances. Such developments did not necessarily require major changes to our design as we had already planned to heavily use AWS and artificial intelligence features. However, due to operating costs necessary to support both dev and prod environments within our AWS services the team had to have a conversation with the sponsor to discuss billing. Ultimately, the sponsor decided that it was necessary to maintain our design in order to support the app's listing on the Shopify marketplace and beta testers.
4. How has the development and implementation progressed? What percentage of the product do you estimate was completed? Is the team providing the documentation within the implementation artifacts?
 - a. By the end of the second semester it is estimated that the entire MVP was 100% completed. This means that if the sponsor so chooses he could realistically distribute this app to Shopify Merchants and they could realize the value that it provides within their shops. However, the app is not fully polished and there may be some functional bugs that may cause user confusion. The app also does not implement some of the bonus features proposed by the sponsor that would have otherwise provided much greater value to the merchants. For example, while our app does provide many meaningful metrics relating to subscription products that Shopify does not natively support, SubscriptionsFlowAI is missing metrics that express how useful the churn guard feature is such as "Number of Saved Subscribers" that would be the amount of times the app prevented a cancellation etc...

5. What was the team's testing strategy? Did the team develop a test plan? If so, was it followed? Did the team perform unit testing? Did the team use any test frameworks, such as JUnit? What are the testing results? Were any major defects found during the system test? If so, were they fixed? Did the team do regression testing?

The team's testing strategy focused on peer review, manual testing, system testing, and regression testing rather than formal unit testing. Once a feature was completed, the developer created a pull request. Other team members then reviewed the code, tested the feature, and provided feedback before the feature was approved and merged. This helped the team catch issues early, confirm that the feature worked as expected, and maintain code quality throughout the project. The team did develop a test plan which included reviewing completed features, testing core workflows, checking the application before sprint demos, and validating fixes after defects were found. The testing results were generally positive. Most features were tested through the pull request process and during system testing. The team also performed regression testing after fixing defects and adding new features. After fixes or updates were made, the team retested related features to confirm that previously working functionality still behaved correctly.

6. Products need to be designed within guidelines and constraints appropriate for each project. It is also important to consider the impacts of the products that are designed. In the following categories discuss the constraints and impacts that have a bearing on your project. Note that all of these categories may not have bearing on your project but your project is probably affected by many of them in ways that you may not think of regularly. probably affected by almost all of these.

i. Economic issues

Constraint: In times of economic hardship our product has a high chance of being used far less as people would be looking to save money, and subscriptions are likely some of the first things that people will stop buying.

In addition, the application will only be deployed to the shopify app store. This means that only merchants who use the Shopify platform can be considered potential customers.

Impact: Improper implementation can lead to wasting company time and money. At the very least our project has to provide value to companies with the subscription management and analytics, if not then that serves as an issue.

ii. Environmental issues

Constraint: If there are environmental issues such as natural disasters, it can bring servers down leading to businesses losing money.

Impact: Utilizing AI and servers to host and run business raises concerns with resources consumptions. Servers can consume large amounts of energy which can have negative impacts on the environment.

Social issues

Impact: Our project can be viewed as slightly anti-consumer since it adds another layer before they can fully cancel their subscription, but at the same time we offer and do provide discounts.

iii. Political issues

Constraint: AI regulations may affect the ChurnGuard AI model.

iv. Ethical issues

Impact: The scope of the project changed from emailing potential lost subscribers before they cancel to intervening when they attempt to cancel. Sending an email to an inactive customer may remind them of a subscription they forgot they had. Thus, the design is taking advantage of people forgetting they are subscribed. Additionally, getting in the way of customers canceling is not great from a consumer perspective.

Constraint: Since the main distinguishing feature of this service involves discounts, some customers may attempt taking advantage of the discount system. We need to prepare for this.

v. Health and safety

N/A

vi. Manufacturability

Constraint: If a product isn't able to be produced quickly, or produced as quickly as the demand requires, then subscriptions for our product become hard to manage/impractical

Impact: Our project can help advertise certain products a merchant has, and if a certain one or many get very popular, the manufacturer may struggle keeping up with demand.

vii. Sustainability

Constraint: Our project is only available if Shopify is available. Even though it is unlikely Shopify goes down, it's always a possibility.

Impact: If the app is not user friendly then admins will have trouble creating and managing subscriptions.

7. What industry or engineering standards was your project required to adhere to? Were these new standards that the team had to learn? Did your sponsor provide you support for understanding these standards? Did you have to educate your sponsor about these standards?
 - a. SubscriptionsFlow AI had to adhere to the following standards
 - i. Built For Shopify
 1. This is a set of guidelines that 3rd party Shopify apps must adhere to in order to qualify for the “Built For Shopify” badge. This badge is a symbol of quality towards merchants and a testimonial from Shopify themselves attesting to the overall satisfaction of the app. Such requirements include properly managing Shopper and User data alike - deleting all data when a customer leaves the store etc... Using Polaris UI components when necessary to achieve a consistent user experience and self hosting your app in a responsible manner.
8. What standard software engineering practices did you follow? Did your sponsor specify any of these practices for you?
 - a. Some software engineering principles that the team followed are below:
 - i. Code Review
 1. As much as possible and as best as we could time permitting the team engaged in insightful code reviews. These events took place mostly asynchronously on discord and included providing feedback on each other’s work to make it stronger. Had we had more time and less to do, the team feels as though dedicated meetings for code review would have been more effective in knowledge sharing
 - ii. Comments/Self-Describing Code
 1. We consistently named functions in ways that described what their main use was, and where and what parameters were being used. Furthermore, if the code seemed confusing, the team would try their best to add comments for clarification, especially in places where lambdas were used due to it being a staple in typescript.

iii. DRY

1. During the winter break and second semester we were able to shift many of our GraphQL calls to a sudo command pattern. We would simply tell an interface what we needed, specifically what data, and pass in the parameters, and it would be able to do the call accordingly. Instead of us having to write a different graphql call for every little different piece of data we needed, we were able to leverage this to remain flexible, and cut down on development time.

iv. Data Minimization

1. One of the possibilities discussed for the ChurnGuard AI was if the AI should have the ability to learn from and improve responses for individual customers in addition to applying the aggregated data from all the customers. We decided against this to ensure we are not collecting personally-identifying information. The data collected for AI training is trimmed to only include customer attributes directly related to churn percentage.

Process and Project

1. What was your process methodology? Was the process appropriate for the project? Did you follow the process or modify it as the project progressed? If you could repeat the project, what would you do differently?
 - a. Our team consistently followed an Agile Scrum methodology by which we progressed through development of our product via bi-weekly sprints. At the start of each sprint we reflected on the previous cycle considering whether or not we achieved our goals for that time. Using this insight we then performed sprint planning, pulling select tickets from the backlog in such a way to maximize sponsor satisfaction and product quality. This method was suitable for our project due to the fact that we had regular communication with our sponsor in order to ensure product quality and relevance. Upon receiving feedback from the sponsor we were able to work effectively during our sprints to continue developing the product iteratively.
 - b. What we may consider doing differently in the future is delaying the actual implementation of the product and instead opting for a waterfall approach in the first semester - to then transition to scrum during development. The product began

similar to a start up with no code and many ideas to be refined. Had we spent more time in the elaboration and inception phases of the product then development may have gone smoother.

2. Was there a large requirement to learn the problem domain? What approach was used to gain domain expertise? Did your sponsor provide adequate support? What forms of support did you receive?
 - a. The team agrees that there was a large requirement to learn and understand the problem domain. None of us had really used Shopify before from the perspective of a merchant so we are wholly unfamiliar with what the system provides compared to what exactly we were providing. The team was also unsure of the means to best develop and deliver features such as subscriptions management. For example, there was a considerable amount of time where we were not sure whether or not Shopify explicitly handles recurring billing in the frame of a subscription contract (of course, they do).
 - b. In order to gain domain knowledge the team spent time doing research via targeted spikes to learn such things as: What data entities and functions does Shopify provide with its GraphQL API? How will merchants use and access the application? How can we develop both shopper frontend and merchant backend features? Etc...
 - c. Initially the team received support from the sponsor in the form of our questions being answered by him. During many of our initial meetings a large portion of the time was spent asking and receiving answers to questions pertaining to not only the project requirements and scope but also to general aspects about Shopify and what it offers to merchants. Our sponsor's preparation prior to beginning this project was instrumental in catching the team up to speed when it came to domain knowledge.
3. What mechanisms did the team use to track project progress? Did they give the team and sponsor adequate insight into project progress and issues? How well did the team track its project progress? How often did these artifacts get updated on the department project sites?

- a. The team mainly used sprint goals and metrics to track progress. At the commencement of the first and second semesters the team spent time creating tickets in the backlog that would ultimately represent the state by which we wanted to deliver the final product. We were thus able to determine our progress by checking up on how many tickets were completed pertaining to each epic. In general however we mostly used sprint velocity to gauge how much work we were completing each iteration.
 - b. In using our Github Projects sprint board the team was adequately able to track our progress. It is unclear whether or not the use of a shared board amongst both the team and sponsor enhanced the visibility of the project for the sponsor though. Updates were primarily shared with the sponsor via spring demos in order to ensure that he could visually ascertain the state of the project and whether or not we were meeting his expectations.
 - c. Progress artifacts were updated on our website on an as-required basis per the schedule for senior project deliverables.
4. Did the team conduct effective meetings?
 - a. In the first semester we may have answered “no” or “not always” to this question. We previously spent large blocks of time during meetings on tangential conversations that although increased our overall confidence and understanding of the project also greatly increased the length of our meetings. To squash this while also maintaining effectiveness of our meetings we more closely followed our meeting agendas which were generally made to be more intentful. This allowed us to continue sharing high volumes of important information while not spending all night in a team room.
5. Did the team meet all project milestones? Which milestones, if any, were missed or were met ahead of schedule? What contributed to schedule changes? What could the team have done differently to ensure that milestones were met?
 - a. It is no secret that we failed to meet the delivery of a beta release project prior to the conclusion of our first semester. This has already been well documented however, we can confidently say that we overpromised to our sponsor last semester. In lieu of our miscalculation we decided to set more realistic yet still satisfactory milestones for this semester. The team agrees that we hit all of such milestones which include: Getting the app ready for production (MVP), finalizing hosting and database environments, and providing highly polished AI features.

Undeniably, our final product has met these goals and we are handing off a solid application that has the potential for further development.

6. Was the team required to adopt new technologies? What were these technologies? What approach did the team use for selecting the appropriate technology for the project? Did the sponsor provide any support for learning these technologies? How well did the team ramp up on the new technologies and begin to apply them effectively?
 - a. The only new technologies the team had to adopt were those that came with the starting shopify polaris app, e.g. prisma db, as well as implementing AI and cloud features. Additionally, the whole team had to struggle with learning how to make queries to the Shopify API. The process wasn't very rigorous for choosing appropriate technologies, as they were mainly laid out beforehand by our sponsor, and already aligned with most of our developer experience.
7. How well did the team maintain quality control over the project artifacts? Have all artifacts been reviewed for adherence to quality standards? What was the review process used by the team?
 - a. Last semester we felt that the only artifact that we did not deliver well was the final presentation. We can proudly say that this semester there were no such shortcomings to speak of as we felt we were more adequately prepared for each deliverable including the presentation. Furthermore, we had a great opportunity to have our technical intern use them as a learning tool of sorts to onboard onto our project, and any gaps they noticed, or areas of improvements that were needed, were documented and improved upon.
8. Did the team have any issues with configuration management? How were these problems solved? What percentage of project artifacts is under configuration control?
 - a. All documents are hosted via google docs which means they are covered in terms of configuration control.
 - b. Now that our dev and prod database are properly hosted and configured running on AWS RDS we no longer have the pain point from last semester regarding schema synchronization issues.
 - c. Overall configuration management has never been a huge issue, and now with all the improvements we made in our process, and specifically the tools we're leveraging, configuration management remains minimal.

9. What was the set of metrics that the team tracked? Did the team gather these metrics on a consistent basis? What did the team learn from the review of these metrics?
 - a. The set of metrics that the team is tracking are sprint velocity, effort by type, SUMI scores, weekly time tracking, and web core vitals. We have been taking note of the process ones effectively, but most of our product metrics require for it to be fully deployed for beta testing which we haven't achieved quite yet. The team is learning that we are scaling well with the project and putting in incredibly high amounts of effort.
 - b. We were able to complete a round of SUMI testing that uncovered some issues with our UI and logical bugs. These problems were documented and adequately addressed. Further iterations of the project could consider to continue testing this metric to ensure that changes made to the app do not negatively impact the overall usability.

Communication and Interaction

1. How well did the team communicate project progress to the sponsor? What regular communication did the team have with the sponsor? Did the team maintain this communication to the satisfaction of the sponsor? Were any adjustments needed in the communication over time? Were these changes initiated by the team or the sponsor?
 - a. The team frequently communicated with the project sponsor to provide updates on the status of the project. Such communication primarily occurred during weekly meetings and focused on high level discussions detailing the sponsor's satisfaction with recently completed changes and the sponsor's desire for the direction of the next sprint. The team also communicated asynchronously via discord to share brief updates on various matters as needed.
 - b. Yes, the team did make adjustments to our communication methods based on lessons learned by the end of the first semester. The team felt that solely using email to communicate small matters such as sharing account information and quick updates to minor features was inefficient and difficult to maintain provenance. To address this, we decided to include our sponsor in our discord channel to make communication more effective and easier to search prior messages to find older information. This change was initiated by the team and well met by the sponsor
2. Did the team need to provide technical input to the sponsor? How well did the team educate the customer in these areas? What mechanism did the team use?
 - a. Yes, often the team needed to provide technical input to the sponsor on issues mainly related to our cloud and AI infrastructures. The sponsor wasn't quite as technical in these areas, but we did our best to convey it in a way that wasn't too technical. The team would draw parallels to real world scenarios if needed or highlight the deficiencies or needs of certain areas if needed in a way that could ground the sponsor in the core of what we wanted to discuss rather than get lost in the context.
3. Was this an effective team? What has been contributing to and detracting from the team's effectiveness? What are the team's weak points? What are the team's strong points? What changes could the team have made to make it more effective?
 - a. Yes, the team feels that overall we were an effective group.

- b. One thing that has contributed to our effectiveness is our undeniable synergy. At each stage in the evolution of our project we efficiently delegated work for each other to do based on the general roles that we each fulfilled. For example, one member naturally fell into being the AI lead due to their prior domain experience. When designing and developing AI features or those adjacent to AI this member made sure to share their knowledge with the team so that we were all able to understand the product and contribute accordingly.
 - c. One thing that may have detracted from our effectiveness has been our lack of experience developing a Shopify app from scratch. We feel that we could have benefitted from longer domain research periods and to have spent more time designing features before implementing them. However, because the team prioritized delivering robust functionality to the sponsor we had to spend less time in the ideation phase of the project which ultimately meant that development suffered from more bugs than expected and ended in a prolonged timeline anyways.
- 4. What mechanism did the team use to communicate with the faculty coach? Was communication with the coach effective? Were there any trouble spots with the faculty coach communications? What could the team or faculty coach have changed to make their communication more effective?
 - a. The team primarily communicated with the faculty coach via weekly meetings and email. We feel that there were no issues with the communication with our faculty coach. That said, we would have liked the senior project website to display due dates more efficiently and for changes to due dates to be communicated more clearly.
- 5. Did the team need to interact with department staff personnel, i.e., the office staff or system administration? Was this handled in a professional manner? Were there any problems with these interactions?
 - a. N/A

6. Does the team have a complete website with all project artifacts stored and up-to-date on the software engineering department webserver? How often were entries on the webserver updated?
 - a. Yes, the team has a website hosted via github pages that displays our up to date project artifacts. Entries were updated on a rolling basis when required to do so.
7. How well has the team made presentations to the sponsor and faculty coach? Was the final project presentation done in a professional manner? Was the poster presentation done in a professional manner? What could have been done to improve the team's presentations?
 - a. Presentations were given to the project sponsor primarily in the form of sprint demos. For the most part, the team was well prepared to adequately showcase new features and improvements. There were likely some times where we had technology difficulties due to local configuration issues but also for reasons out of our control such as cloudflare and AWS outages. Overall, these demos effectively communicated changes to the project and the sponsor has seemed satisfied.
 - b. The poster presentation was given in a professional manner and was well rehearsed. The team spent at least 2-3 hours practicing the presentation and even more so creating the slide deck.
 - c. The poster presentation was also professionally given however the team feels that the environment and timing was not ideal to be properly judged by the IAB. Having only been given about 10 minutes to explain our project to 3 separate members in a noisy, crowded room we were not able to communicate well. For example, one IAB member was nearly unable to hear us and we felt it inappropriate to have to raise our voices any louder for fear of coming off rude. Additionally, since each IAB member needed to visit many groups they did not spend enough time with us to be able to gain a full understanding of our project.
 - d. In order to improve the poster presentations we would have liked to have more time and a less noisy space in which to present.
8. Does the technical report adequately document the project and its results? Was the paper of high technical and editorial (language, style, grammar, etc.) quality? Did all teammates contribute to the paper? Did the sponsor contribute to the paper? Did the sponsor review the paper?

The technical report adequately documents the project in both processes, from start to finish, results, and specifics from architecture, to milestones, etc. Everyone contributed a

significant portion and were able to provide input on aspects that they felt they contributed a lot of effort towards. The sponsor did not contribute to or review the paper just yet, but will be granted access once finished. Overall the paper is of high quality, and greatly reflects our project in its entirety.

Preparation and Resources

1. Did the team possess adequate management and process skills (team building, planning, risk management, change management, process definition and tracking, etc.) to carry out the project? If not, how could the program provide better preparation?
 - a. Yes, the team consistently was able to manage all of the items above. The greatest show of this was our ability to onboard a new recruit in a short period of time and allow them to contribute to the project in ways that they both found interesting, and we needed. We also consistently sought out improvements in our process and consistently communicated any concerns or changes that may affect others. Our process had noticeable improvements from the previous semester, and process definition and tracking both became more accurate as our domain understanding grew. All of this ensured that surprises were kept to a minimum, and we could put our entire focus on the task at hand, and not have to worry about what impacts our work now may have down the road.

2. Did the team possess adequate technical skills (requirements, design, coding, testing, quality reviews, etc.) to carry out the project? If not, how could the program provide better preparation?
 - a. The team felt adequately prepared to tackle this project based on our academic and professional experiences in the aforementioned skills. However, not many classes give opportunities for students to deploy completed projects to production environments, a skill that is almost certain to be used in any software industry. Additionally, there are not a lot of classes that teach students to create and manage numerous environments for development, testing and production (Dev Ops). Each member having such skills would have certainly made the deployment process much less daunting. Furthermore, having a class that focuses on bridging the communication gap between technical and nontechnical, or at least a class that has that as one of its main focuses would be incredibly helpful. In the real world this is a common issue that many of us have seen from even older engineers, and being able to develop that skill early would help not only in this class, but throughout one's career.

3. What technical resources (or skills, training, tools) were missing, if any, that would have helped make the senior project experience more successful?
 - a. Exposure to using third party APIs to provide functionality (LLM, Shopify, etc...). Familiarity with the cloud should be in more classes rather than just a small subset. Similar to AI, it's incredibly prevalent in industry, and if students unfortunately don't get the ability to, it could be a detriment to their experience and growth.
4. Were the facilities adequate for you to perform your work on the project?
 - a. Yes, if anything was ever needed from the school or coach, it was provided. Our needs for this project were very minimal though, so not much was required from either party.

Achieving Customer Satisfaction

1. In the team's opinion did the work satisfy the project sponsor? Are there areas where you think you exceeded the sponsor's expectations? Were there any weak spots in this regard?

In the team's opinion, the work satisfied the project sponsor. Throughout the project, the sponsor consistently provided positive feedback and appeared satisfied with the team's progress, communication, and the direction of the application. The team made an effort to keep the sponsor involved through regular meetings, demos, and communication through email and Discord. This allowed the sponsor to provide feedback throughout development and helped the team stay aligned with the sponsor's expectations. The team believes it exceeded the sponsor's expectations in several areas. One area was usability. The team made a strong effort to create an application that was user-friendly and aligned with the sponsor's goal of making the product easy to use. Another area was technical effort. The team spent significant time researching the Shopify API, implementing the core features, preparing the application for Shopify's review process, and hosting the application on its own AWS server. One weak spot was that the team was not able to complete the MVP during the first semester as originally planned. The delay was mainly due to the complexity of the project and the amount of research required to understand the Shopify API. However, the team responded by increasing effort during later sprints, especially during Sprints 4 through 6, when all team members contributed more than 12 hours per week to complete the bulk of the features.

Achieving Team Satisfaction

1. Did the project satisfy the team's expectations for learning? What could the team have done differently to improve the team's learning experience? What could the faculty coach and department have done differently to improve the team's learning experience?

Yes, the project satisfied the team's expectations for learning. Throughout the project, every team member learned new technical and professional skills. At the beginning of the project, no one on the team was familiar with the Shopify API, so there was a significant learning curve. Through research, experimentation, and implementation, the team became more comfortable using the Shopify API and understanding how it connects with the rest of the application. This gave the team valuable experience working with a real third-party platform and helped everyone grow technically. One thing the team could have done differently to improve the learning experience was to plan more thoroughly at the beginning of the project and break tickets down into smaller, more manageable tasks. Many of the tickets were written at an epic level, which made it harder to estimate effort and track progress. Smaller tickets would have made the work clearer and helped the team identify blockers earlier. The team also could have shared research findings more formally, such as through short technical notes, documentation, or internal demos, so that knowledge was spread evenly across all team members.

Advice for Future Teams

1. What one piece, or more, of advice would you give to future senior project teams to help them be successful?

One piece of advice we would give to future senior project teams is to plan carefully and thoroughly, but also remain flexible when changes happen. A strong plan helps the team understand the project scope, divide responsibilities, estimate work more accurately, and stay organized throughout the semester.